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COLLEGE OF ENGINEERING & MANAGEMENT

(An Autonomous Institution)

Adyar, Mangaluru

Department of Computer Science & Engineering

Seminar On
Self-Healing Cybersecurity Systems Using Reinforcement Learning

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Abstract

Modern cybersecurity infrastructures face rapidly evolving threats such as zero-day attacks, ransomware, advanced persistent threats, and multi-stage intrusions. Traditional defense mechanisms rely heavily on static rules, signature databases, and manual intervention, which limits their ability to respond quickly and adapt to unknown attack patterns. This delay often increases system downtime, financial loss, and security risks. To overcome these limitations, self-healing cybersecurity systems have emerged as an intelligent and automated defense paradigm capable of detecting, responding to, and recovering from incidents without human involvement. This report investigates the use of Reinforcement Learning (RL) to design adaptive and self-healing cybersecurity frameworks. RL enables an intelligent agent to learn optimal defensive actions through continuous interaction with the environment by maximizing cumulative rewards. The system automatically selects actions such as blocking malicious traffic, isolating compromised hosts, updating policies, and restoring services. Three recent research works are analyzed: an adaptive RL-based incident response framework that optimizes defense strategies, an RL-driven malware investigation model that accelerates forensic analysis, and a deep RL approach for automating post-alert decision making in SIEM systems. The findings indicate that RL-based approaches significantly reduce response time, improve detection accuracy, lower false positives, and enhance overall resilience compared to rule-based methods. Hence, reinforcement learning provides a promising foundation for building intelligent, autonomous, and self-healing cybersecurity systems for modern digital infrastructures.

Introduction

- Growth of cloud computing, IoT devices, and large-scale networks has significantly increased connectivity, where each connected device becomes a potential entry point for attackers, thereby expanding the overall attack surface and making systems more vulnerable and complex to secure.
- Traditional cybersecurity methods rely mainly on signature-based and rule-based detection techniques, which are effective only for known threats and fail to identify zero-day attacks, advanced persistent threats, and evolving attack patterns.
- Manual incident response processes are slow, time-consuming, and prone to human errors, making them inefficient in handling large-scale or real-time cyber threats, especially in complex and distributed environments.
- There is a growing need for intelligent, adaptive, and automated cybersecurity systems that can monitor continuously, detect threats in real time, and respond quickly without heavy human intervention.
- Reinforcement learning enables systems to learn from dynamic environments through continuous interaction, allowing them to adapt to new threats, optimize response strategies, and improve performance over time.
- Such intelligent systems support the concept of self-healing cybersecurity, where the system can automatically detect attacks, take appropriate actions, recover from damage, and enhance its future responses based on past experiences.

Problem Statement

Problem Statement : The increasing sophistication and frequency of cyber attacks demand intelligent, adaptive, and autonomous defense mechanisms. However, existing cybersecurity solutions largely rely on static rule-based systems or manual intervention, which are slow to respond and ineffective against evolving threats. This highlights the need for a scalable and selfhealing cybersecurity framework that can automatically detect, respond to, and recover from attacks using reinforcement learning–based decision-making.

Objectives:

- To analyze existing Neural Architecture Search (NAS) techniques and identify their limitations in terms of computational cost and scalability.
- To design a training-free Neural Architecture Search framework that eliminates repeated model training during the architecture search phase.
- To integrate evolutionary algorithms for efficient exploration and exploitation of the neural architecture search space.
- To develop a scalable and resource-efficient NAS solution suitable for real-world and resource-constrained environments

Methodology and Implementation



Figure 1. Architecture diagram for the proposed self-healing cybersecurity framework

- The monitoring system continuously collects real-time data such as network traffic, system logs, and user activities from different sources.
- The collected data is processed in the preprocessing stage where noise is removed and the data is converted into a structured format for analysis.
- A trained model is developed using a dataset to learn patterns of normal and malicious behavior in the system.
- The intrusion recognition module uses this trained model to analyze incoming data and detect any suspicious or abnormal activities.
- If an intrusion is detected, the system generates an alert report, which can be used to notify administrators or trigger automatic response actions.

Outcome of Seminar

- Developed a clear understanding of reinforcement learning concepts and their application in cybersecurity.
- Identified the limitations of traditional rule-based and manual security systems.
- Explored various existing RL-based approaches for threat detection and response.
- Designed a conceptual self-healing cybersecurity framework integrating detection, response, and recovery.
- Understood the importance of automated and adaptive defense mechanisms in modern cyber environments.
- Analyzed challenges such as scalability, computational cost, and safe decisionmaking.
- Gained insights into real-world applicability of autonomous cybersecurity systems.
- Improved knowledge in designing intelligent and resilient security solutions.

Sustainable Development Goals Mapping

SDG Number	SDG Name	Contribution to the Proposed Work
SDG 9	Industry, Innovation and Infrastructure	Enhances digital infrastructure security using reinforcement learning. Promotes innovation through adaptive and automated cybersecurity solutions.
SDG 16	Peace, Justice and Strong Institutions	Strengthens protection of critical systems and sensitive data. Helps reduce cybercrime and ensures secure digital operations.
SDG 8	Decent Work and Economic Growth	Reduces downtime and financial losses caused by cyber attacks. Improves productivity through automated and efficient security

Table 2. Mapping of Seminar Topic to Sustainable Development Goals (SDGs)

Conclusion

This work presented an efficient self-healing cybersecurity framework based on reinforcement learning for automated threat detection, response, and recovery. By modeling cybersecurity defense as a sequential decision-making problem, the proposed approach enables autonomous and adaptive mitigation strategies without relying heavily on manual intervention or static rule-based mechanisms. The integration of intelligent monitoring, reinforcement learning–based policy optimization, and automated response mechanisms significantly reduces response time, operational overhead, and human dependency while improving overall system resilience.

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